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Practical 3

3. Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution. Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as `ps`, `top`, and `/proc`. Document the observations

Execution:

```
jamuna@Jem:~$ nano practical3.c
jamuna@Jem:~$ gcc practical3.c
jamuna@Jem:~$ ./a.out
Read end of the pipe is 3
Write end of the pipe is 4
Pipe successfully created
Parent process
The process ID is 2377
The parent process ID is 2318
Child process
The process ID is 2378
The parent process ID is 2315
```

```
jamuna@Jem:~$ ps
  PID TTY          TIME CMD
 2318 pts/0        00:00 bash
 2420 pts/0        00:00 ps
```

```

jamuna@Jem:~$ top
top - 05:20:01 up 1:54, 1 user, load average: 0.03, 0.01, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.6 us, 0.0 sy, 0.0 ni, 98.7 id, 0.0 wa, 0.0 hi, 0.6 si, 0.0 st
MiB Mem : 7762.6 total, 7102.5 free, 566.6 used, 246.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7196.1 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24140   15400   11604 S   0.0   0.2   0:01.21 systemd
    2 root        20   0    3180    2212    2072 S   0.0   0.0   0:00.02 init-systemd(Ub
    6 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00 init
   49 root        19  -1   50400   16840   15548 S   0.0   0.2   0:00.41 systemd-journal
   81 systemd+    20   0   22416   14484   11984 S   0.0   0.2   0:00.11 systemd-resolve
   87 root        20   0   35364   12328    9240 S   0.0   0.2   0:05.74 systemd-udev
  107 root        20   0    2888    1948    1820 S   0.0   0.0   0:00.07 chronyd-starter
  108 root        20   0    4472    3140    2892 S   0.0   0.0   0:00.03 cron
  109 message+   20   0    8820    5476    4772 S   0.0   0.1   0:00.12 dbus-daemon
  115 root        20   0   44096   29892   14752 S   0.0   0.4   0:00.31 networkd-dispat
  150 root        20   0   18436    9412    8168 S   0.0   0.1   0:00.12 systemd-logind
  177 syslog      20   0  220548    5936    4660 S   0.0   0.1   0:00.19 rsyslogd
  218 root        20   0    5492    2776    2588 S   0.0   0.0   0:00.01agetty
  232 _chrony      20   0   23844   11376    7224 S   0.0   0.1   0:00.25 chronyd
  240 _chrony      20   0   12096    2436    1796 S   0.0   0.0   0:00.02 chronyd
  273 root        20   0  123460   32464   17856 S   0.0   0.4   0:00.18 unattended-upgr
  334 root        20   0    8648    5584    4740 S   0.0   0.1   0:00.02 login
  379 jamuna     20   0   22336   13512   10888 S   0.0   0.2   0:00.24 systemd
  381 jamuna     20   0   22912    4088    2088 S   0.0   0.1   0:00.00 (sd-pam)
  411 jamuna     20   0    6136    5436    3864 S   0.0   0.1   0:00.07 bash
 2137 polkitd     20   0  306444    8356    7544 S   0.0   0.1   0:00.12 polkitd
 2313 root        20   0    3184    1124     980 S   0.0   0.0   0:00.00 SessionLeader
 2315 root        20   0    3200    1260    1108 S   0.0   0.0   0:00.04 Relay(2318)
 2318 jamuna     20   0    6268    5544    3872 S   0.0   0.1   0:00.11 bash

```

```

jamuna@Jem:~$ /proc
-bash: /proc: Is a directory

```